

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	10 February 2026
Team ID	LTVIP2026TMIDS82993
Project Name	HouseHunt: Finding Your Perfect Rental Home
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration	• Registration through web form (Name, Email, Password) • Role selection (Admin / Owner / Renter) • Password encryption using bcrypt • Store user details in MongoDB
FR-2	User Login & Authentication	• Login using Email and Password • Password verification using bcrypt • JWT token generation • Token-based session management • Role-based access control
FR-3	Role-Based Authorization	• Admin access control • Owner access control • Renter access control • Protected routes using middleware • Unauthorized access prevention
FR-4	Property Management (Owner Module)	• Add new property • Update property details • Delete property • View own properties • Track property availability status
FR-5	Property Viewing (Renter Module)	• View all available properties • View property details • Filter properties (future enhancement) • View rent amount and address
FR-6	Booking Management	• Renter can book a property • Owner can view bookings • Track booking status • Update property availability upon booking
FR-7	Admin Monitoring	• View all users • View all properties • Monitor bookings • Manage system-level access

FR-8	API Integration	• REST API endpoints • Axios integration in frontend • JSON data exchange • Proper HTTP status codes
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Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	• Simple and user-friendly interface • Clear navigation between dashboards • Responsive design • Easy property listing and booking process
NFR-2	Security	• Password hashing using bcrypt • JWT-based authentication • Role-based authorization middleware • Protected API endpoints • CORS configuration
NFR-3	Reliability	• Proper error handling in backend • Validation of user inputs • Consistent database operations • Secure API response handling
NFR-4	Performance	• Lightweight REST APIs • Asynchronous request handling • Efficient MongoDB queries • Minimal server response time
NFR-5	Availability	• Local server availability during development • Can be deployed to cloud (AWS / Render / Vercel) • 24/7 accessibility after deployment
NFR-6	Scalability	• 3-tier architecture • Modular backend structure • Easy to add new features • Cloud deployment support • Can scale horizontally with load balancers (future scope)